



Arjun Nagar

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B.Tech – Electrical Engineering

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Portfolio

EDUCATION

•National Institute Of Technology, Kurukshetra

B.Tech. in Electrical Engineering

2023 – 2027

CGPA: 7.71

•Modern Vidya Niketan Senior Secondary School

CBSE (Class XII), Faridabad

2022

87.4%

PERSONAL PROJECTS

•Schedura – AI-Powered Social Media Automation Platform

React.js, Node.js, Express.js, MongoDB, Gemini API, Leonardo AI, JWT, Cloudinary

Live

GitHub

- Developed a full-stack social media automation platform using the **MERN** stack with secure **JWT-based authentication** and role-based access control (RBAC).
- Integrated **Gemini API** and **Leonardo AI** to automatically generate social media captions and AI-created images, streamlining content creation workflows.
- Built a post scheduling system with multi-platform publishing support and integrated **Cloudinary** for efficient media upload, storage, and delivery.
- Designed an analytics dashboard to visualize engagement metrics and post performance, enabling users to monitor campaign effectiveness through real-time insights.

•KineticCare – AI-Powered Physiotherapy Platform

React.js, MediaPipe BlazePose, Three.js, Node.js, MongoDB, Tailwind CSS

Live

GitHub

- Developed a browser-based physiotherapy platform that performs real-time human pose estimation using **MediaPipe BlazePose** with **Three.js**-based 3D skeletal visualization.
- Implemented **3D vector mathematics** and joint-angle calculations to measure range of motion, achieving approximately $\pm 2.3^\circ$ **mean absolute error** for exercise tracking.
- Built an automated exercise evaluation system using a **finite-state (hysteresis) machine** to count repetitions, validate posture, and reduce false-positive detections.
- Integrated browser-native **Web Speech API** and **Web Audio API** to provide low-latency voice guidance and real-time corrective feedback during rehabilitation exercises.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, JavaScript

Frontend: HTML5, CSS3, React.js, Next.js, Tailwind CSS, Redux Toolkit, React Router, Responsive Web Design

Backend: Node.js, Express.js, RESTful APIs, JWT Authentication, OAuth 2.0, Bcrypt.js, Multer

Databases: MongoDB, PostgreSQL

Cloud & DevOps: AWS S3, AWS SDK, Docker, Kubernetes, Git, GitHub, CI/CD, Vercel, Postman

Core Subjects: DSA, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering, System Design Fundamentals

POSITIONS OF RESPONSIBILITY

•Active Member, National Service Scheme (NSS), NIT Kurukshetra

2023 – Present

- Collaborated across teams to coordinate institute outreach events, strengthening teamwork and communication.

•Valorant Esports Player, NIT Kurukshetra Esports Team

2024 – Present

- Represented the institute in competitive Valorant scrims and esports tournaments.

ACHIEVEMENTS

•Led a project team to achieve **14th place** in the **VISIONARY Hackathon 2.0 – Mettl Xathon** by developing **KineticCare**, a physiotherapy platform featuring **3D kinematic motion tracking**.

2025

•Maintained an active competitive programming profile with aggregated performance across coding platforms on **Codolio**, including solving **500+ DSA problems**.

Ongoing